

Semi-Supervised Multimodal Emotion Recognition with Class-Balanced Pseudo-Labeling

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ABSTRACT

This paper presents our solution for the Semi-Supervised Multimodal Emotion Recognition Challenge (MER2023-SEMI), addressing the issue of limited annotated data in emotion recognition. Recently, the self-training-based Semi-Supervised Learning (SSL) method has demonstrated its effectiveness in various tasks, including emotion recognition. However, previous studies focused on reducing the confirmation bias of data without adequately considering the issue of data imbalance, which is of great importance in emotion recognition. Additionally, previous methods have primarily focused on unimodal tasks and have not considered the inherent multimodal information in emotion recognition tasks. We propose a simple yet effective semi-supervised multimodal emotion recognition method to address the above issues. We assume that the pseudo-labeled samples with consistent results across unimodal and multimodal classifiers have a more negligible confirmation bias. Based on this assumption, we suggest using a class-balanced strategy to select top-k high-confidence pseudo-labeled samples from each class. The proposed method is validated to be effective on the MER2023-SEMI Grand Challenge, with the weighted F1 score reaching 88.53% on the test set.

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CCS CONCEPTS

• **Computing methodologies** → **Computer vision; Natural language processing.**

KEYWORDS

multimodal emotion recognition; semi-supervised learning; self-training; pseudo-labeling; class imbalance

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1 INTRODUCTION

Automatic multimodal emotion recognition has attracted significant attention for its vital role in human-computer interactions. However, annotating emotions is time consuming and expensive, making the existing emotion-related datasets limited in scale. Therefore, employing large amounts of unlabeled data for semi-supervised multimodal emotion recognition is crucial. The Semi-Supervised Multimodal Emotion Recognition Challenge (MER2023-SEMI) [15] provides a comprehensive dataset for researchers to explore Semi-Supervised Learning (SSL) methods.

Self-training based SSL methods have yielded promising results across various tasks [7, 28, 29]. These methods leverage unlabeled data by generating pseudo-labels and use labeled and pseudo-labeled data as ground truth during training. Despite the effectiveness of self-training, it is susceptible to erroneous pseudo-labeled data. This is because the model's prediction confidence of erroneous pseudo-labeled data increases over time during training, making it incapable of correcting its own mistakes. Consequently, errors accumulate progressively throughout the training process.

To reduce the confirmation bias of the pseudo-labeled data, some works [13] adopted high thresholds to select the unlabeled data with high confidence. However, a constant threshold can lead to the imbalance of pseudo-labeled data class. For example, “happiness” occurs more often in emotion recognition than other emotions. Therefore, the trained model tends to assign higher confidence to the classification results of “happiness”, which causes the risk of selecting an imbalanced subset predominantly labeled as “happiness” when employing a constant threshold. Therefore, in self-training-based multimodal emotion recognition, the critical challenges revolve around effectively selecting the correct pseudo-labeled data while mitigating the class imbalance concerns.

Recently, several works [2, 18, 35] have proposed curriculum pseudo-labeling, which selects pseudo-labeled data in a progressive manner, starting from easier samples and gradually moving towards challenging ones. Among them, the work in [35] suggests assigning dynamic thresholds to different classes based on prediction difficulty, effectively mitigating the potential class imbalance issue. Inspired by these works, we propose a simple yet effective semi-supervised multimodal emotion recognition method, where the core strategy is to select the high-confidence pseudo-labeled samples from each class in a progressive manner. Unlike the previous studies focusing on tasks involving a single modality, this paper extensively utilizes the multimodal information inherent in emotion recognition tasks to minimize further the confirmation bias in selecting the pseudo-labeled data. Moreover, we propose to utilize the class-balanced top-k selection strategy as an alternative to the dynamic thresholding methods, aiming at reducing the class imbalance in selecting the pseudo-labeled data. The proposed approach not only avoids the complexity of dynamically setting the threshold to reduce confirmation bias, but also mitigates the issue of class imbalance when selecting pseudo-labeled data.

In general, this paper has the following contributions:

- A self-training-based semi-supervised framework is proposed to leverage the labeled and unlabeled data for multimodal emotion recognition.
- A novel class-balanced top-k pseudo-labeling strategy with multimodal cues is proposed to reduce confirmation bias and alleviate class imbalance issue of the pseudo labeled data.
- The proposed method achieves good performance in MER2023-SEMI grand challenge, demonstrating its effectiveness for semi-supervised multimodal emotion recognition task.

2 RELATED WORK

Multimodal Emotion Recognition: In recent years, various methods have been proposed to exploit the cross-modal relationships for multimodal emotion recognition [3, 25, 26]. [26] utilizes a Cross-modal Transformer to fuse any two modalities, [38] employs Cascade Multi-Head Attention for multimodal fusion. The method of [20] introduces progressive modality reinforcement, which incorporates a message hub to exchange complementary information among different modalities. In addition to transformers [5], some approaches [10, 14, 19, 34] convert multimodal features into graph-structured data and utilize graph neural networks to learn multimodal representations, yielding promising results. Moreover,

the work of [22] integrates contrastive learning with multimodal fusion to learn more discriminative representations.

Semi-Supervised Learning: SSL methods leverage large amounts of unlabeled data to address the issue of limited annotated data. Recently, [13, 23] employed a pre-defined constant threshold to select pseudo-labeled samples. While this strategy reduces the confirmation bias by selecting high-confidence unlabeled data, it overlooks large amounts of other unlabeled data in the early training stages. To address this issue, [30] selects more informative unlabeled samples by dynamically adjusting the threshold. Similarly, [35] proposes Curriculum Pseudo-Labeling (CPL), which flexibly adjusts thresholds for different classes to select pseudo-labeled data.

Self-training: Self-training is a form of SSL where a model is initially trained on labeled data and then extended with pseudo-labeled data. The work of [39] applies self-training on speech emotion recognition and achieves state-of-the-art performance. The approach of [21] utilizes a progressive correction of segment-level labels, which improves the performance of utterance-level speech emotion recognition. The work of [11] introduces an adaptive inter-class confidence adjusting strategy for facial expression recognition tasks. Similarly, the work of [8] assigns dynamic weights to each pseudo-labeled sample based on the training loss.

3 METHOD

3.1 Problem Formulation

MER2023-SEMI provides both labeled data $\tilde{D}_l = \{(x_i, e_i, v_i)\}_{i=1}^{N_l}$ and unlabeled data $\tilde{D}_u = \{\tilde{x}_i\}_{i=1}^{N_u}$, where $e_i \in \{1, 2, \dots, C\}$ and $v_i \in [-5, 5]$ represent the label of discrete emotion and dimensional emotion of the sample x_i . Here, N_l, N_u, C are the number of labeled, unlabeled, and emotion categories, respectively. For each sample x_i (or $\tilde{x}_i$), we extract visual features $f_i^v \in \mathbb{R}^{t_v \times d_v}$, acoustic features $f_i^a \in \mathbb{R}^{t_a \times d_a}$ and lexical features $f_i^l \in \mathbb{R}^{t_l \times d_l}$, where d_v, d_a, d_l denote the feature dimension, and t_v, t_a, t_l denote the sequence length. For the self-training SSL, we want to select a subset of the unlabeled data $\tilde{D}_u^p \in \tilde{D}_u$ with accurate pseudo-labels, denoted as $\tilde{D}_u^p = \{\tilde{x}_i, \tilde{e}_i, \tilde{v}_i\}_{i=1}^{N_u^p}$, $N_u^p < N_u$, and combine them with the labeled data $\tilde{D}_u^p \cup \tilde{D}_l$ to train the model for boosting its performance.

3.2 Multi-modal Network Architecture

Figure 1 shows the proposed multimodal emotion recognition model. **Feature Extraction:** The feature sequences of visual(f^v), lexical(f^l) and acoustic(f^a) modalities are extracted.

- Visual: We use ResNet50 pre-trained on BP4D [4, 36, 37] to extract 512-dimensional features for each face image. The feature sequence f^v is normalized with a window length of 0.2s and a hop size of 0.2s, the mean feature in each window is adopted.

- Acoustic: We utilize HuBERT [9] to extract 1024-dimensional frame-level speech features. Then, we pre-process the feature sequence f^a as in the case of the visual modality.

- Lexical: We utilize MacBERT [6] to extract 1024-dimensional features for each word and obtain the feature sequence f^l .

Feature Encoding: A fully-connected layer is first used to map each time step's features to d dimension for all the modalities. Considering the features at different time steps may contribute differently to emotion recognition, we employ the attention mechanism

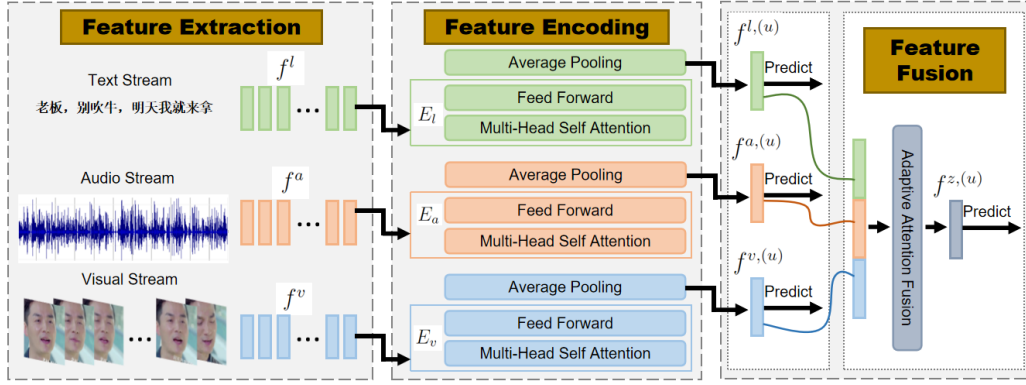


Figure 1: The framework of the multimodal emotion recognition model.

based Transformer [17, 27, 31] to encode the feature sequences of different modalities and obtain the utterance-level features of each modality by average pooling the encoded feature sequences. The overall process can be described as follows:

$$f^{m,(u)} = \text{Pooling}(E_m(f^m, \theta_m)), m \in \{l, v, a\} \quad (1)$$

where $f^m \in \mathbb{R}^{t_m \times d_m}$ is the frame-level feature sequence, $f^{m,(u)} \in \mathbb{R}^d$ is the utterance-level feature. E_m denotes the combination of fully-connected layer and transformer encoder with trainable parameters θ_m , Pooling is the temporal average pooling function.

Multimodal Fusion: As discussed in [15], we calculate an importance score for each modality and apply weighted fusion to generate the multimodal features, as follows:

$$h^m = \text{Concat}(h^{l,(u)}, h^{a,(u)}, h^{v,(u)}), w = \text{Softmax}(\text{FFN}(h^m, \theta_w)) \quad (2)$$

Where FFN is the fully connected layers with parameter θ_w . The attention scores $w \in \mathbb{R}^{3 \times 1}$ are calculated over the concatenated features $h^m \in \mathbb{R}^{d \times 3}$, the multimodal feature $f^{z,(u)}$ is obtained through the attention-weighted fusion.

3.3 Loss Function

As shown in Figure 1, we utilize both unimodal features and multimodal fused features $f^{m,(u)}, m \in \{l, v, a, z\}$ to predict discrete emotion $\hat{e}^m$ and valence $\hat{v}^m, m \in \{l, v, a, z\}$. Due to inconsistencies between unimodal and multimodal labels for certain samples [16, 33], we choose to train the unimodal prediction head on multimodal annotations without backpropagating the gradients. During the training process, we employ a multi-task learning objective, where the cross-entropy loss $\mathcal{L}_e$ is used for discrete emotions and the mean squared error (MSE) $\mathcal{L}_v$ is used for valence, as follows:

$$\mathcal{L}^m = \mathcal{L}_e^m + \lambda \mathcal{L}_v^m$$

$$\mathcal{L}_e^m = -\frac{1}{N_l + N_{u,(p)}} \sum_{i=1}^{N_l + N_{u,(p)}} e_i \log(\hat{e}_i^m), \mathcal{L}_v^m = \frac{1}{N_l} \sum_{i=1}^{N_l} (v_i - \hat{v}_i^m)^2 \quad (3)$$

where λ is a weighting factor to balance two different loss functions, the final loss is obtained by averaging the losses $\mathcal{L}^m, m \in \{l, v, a, z\}$ from different uni-modalities and multi-modality.

3.4 Self Training

Self-Training Process: We first train an initial multimodal emotion recognition model (see Figure 1) using the labeled data, then repeat the self-training process by augmenting the labeled data with pseudo-labeled data to retrain the model, until the multimodal consistency and top-k strategies could not be satisfied, or a maximum number of iterations is reached. For the pseudo-labeled data, we only select the top-k predictions of each class from the unlabeled data, based on the multimodal consistency.

It is noted that in MER2023-SEMI, only the classification performance of the discrete emotions is evaluated. Therefore, in the training process of the initial model with the labeled data, both the discrete emotion classification loss and the regression loss of valence are calculated, as shown in Equation (3). But in the self-training process, only the discrete emotion classification task is taken into account for the pseudo-labelled samples.

Algorithm 1 The self-training process

Input labeled & unlabeled data $\tilde{D}_l = \{(x_i, e_i, v_i)\}_{i=1}^{N_l}, \tilde{D}_u = \{\tilde{x}_i\}_{i=1}^{N_u}$
Output the multimodal emotion recognition model Φ
initialization
 Train the model Φ using the labeled data $\tilde{D}_l$.
repeat
 1. Use model Φ to predict the labels of remaining unlabeled data;
 2. Select a subset of pseudo labeled data $\tilde{D}_u^p$ following multimodal prediction consistency and balanced top-k selection strategy;
 3. Retrain the model Φ with $\tilde{D}_u^p \cup \tilde{D}_l$;
until convergence or the maximum number of iterations is reached

Multimodal Prediction Consistency: We propose to utilize multimodal information for further reducing the confirmation bias. We assume that the classification confidence would be reliable when the predicted outcomes from the unimodal and the multimodal are identical. Therefore, we consider a pseudo-labeled sample trustworthy and add it for training only when the predicted emotion categories from the unimodal and multimodal branches are identical.

Balanced Top-K pseudo-labeling: To reduce the confirmation bias while preventing class imbalance, we design a simple yet efficient class-balanced pseudo-labeling strategy by selecting top-k confidence unlabeled samples from each class. Compared to the

Table 1: Supervised learning recognition results with five-fold cross-validation. ‘Dis’, ‘Dim’ and ‘Com’ denote the performance on discrete, dimensional and combined metrics.

Modality	Method	Train & Val			Test Dis(↑)
		Dis(↑)	Dim(↓)	Com(↑)	
Acoustic	Baseline	65.67	1.27	0.34	85.11
	Ours	66.15	1.26	0.35	-
Lexical	Baseline	42.62	2.39	-0.17	46.32
	Ours	46.50	2.20	-0.09	-
Visual	Baseline	57.48	1.38	0.23	58.01
	Ours	60.58	1.48	0.24	-
Acoustic & Lexical	Baseline	67.02	1.16	0.38	85.59
	Ours	67.82	1.13	0.40	86.47
Acoustic & Visual	Baseline	72.92	0.86	0.52	86.75
	Ours	73.69	0.84	0.53	87.12
Lexical & Visual	Baseline	61.19	1.28	0.29	66.96
	Ours	65.88	1.03	0.40	74.11
Acoustic&Visual&Lexical	Baseline	72.64	0.83	0.52	86.40
	Ours	72.57	0.85	0.51	86.98

commonly used dynamic threshold methods, the proposed method avoids the complex calculation processes of dynamic threshold while ensuring the balanced categories of pseudo-labeled samples.

4 EXPERIMENTS AND RESULTS

4.1 Dataset and Implementation Details

MER2023-SEMI provides two subsets Train&Val and Test with 3373 and 73982 samples, respectively. The Train&Val set is labeled with discrete emotions and valence. Since training/validation splits are not provided, we use five-fold cross-validation for hyperparameters tuning [15]. The official metrics weighted average F-score and MSE are used to evaluate the performance of discrete and dimensional emotions, respectively. It is noted that for MER2023-SEMI, only the classification performance of discrete emotion is evaluated.

For pre-processing the videos, the open-source toolkit OpenFace [1] detects and aligns the faces into 128×128 pixels. For the lexical modality, we obtain the transcripts through the open-source toolkit WeNet [32]. The extracted features of different modalities are mapped to $d = 256$ dimension. The transformer encoders $E_m, m \in \{l, v, a\}$ are with 2 layers and 4 heads. In the self-training process, we select top $k = 10$ pseudo-labeled samples from each class and set the maximum number of steps $N_r = 10$. In each step, we set the maximum number of epochs to 30 and the batch size to 32. The Adam [12] optimizer is utilized with a learning rate of $3e - 4$. Dropout [24] is also exploited with a rate of 0.3.

4.2 Comparison Under Supervised Learning

The experiments are first conducted on training set with supervised learning. The results are shown in Table 1. We use different visual features but the same acoustic and lexical features as Baseline [15].

Table 2: Multimodal self-training results of five-fold cross-validation. ‘Dis’, ‘Dim’ and ‘Com’ denote the performance on discrete, dimensional and combined metrics.

Modality	Iters (N_r)	Train&Val			Test Dis (↑)
		Dis (↑)	Dim (↓)	Com (↑)	
Acoustic & Visual	$N_r = 0$	73.69	0.84	0.53	87.12
	$N_r = 5$	74.74	0.77	0.55	87.75
	$N_r = 10$	75.78	0.75	0.57	88.53

Unimodal Recognition Results: As can be seen in Table 1, the performance of acoustic modality is superior to the other modalities. In addition, compared to the method proposed in the baseline paper [15], employing an additional transformer encoder to encode further frame-level features yields slightly improved results for both acoustic and linguistic modalities. This observation implies that employing attention-based models to emphasize the key frames in the temporal sequence benefits emotion recognition.

Multimodal Recognition Results: Table 1 also presents the performance of the combination of different modalities. One can notice that multimodal fusion consistently outperforms unimodal features by exploring the complementary information from different modalities. The best recognition performance is obtained by using the acoustic and visual modalities, which are used into the self-training process to boost performance.

4.3 Multimodal Emotion Recognition Results with Self-training

Table 2 shows the results of multimodal self-training with different numbers of iterations. As can be seen, exploiting the self-training strategy could boost the emotion classification performance on both validation set and test set, demonstrating that the proposed method could effectively select appropriate pseudo-labeled data from the unlabeled samples, and leveraging pseudo-labeled data for model training can effectively improve the recognition performance.

5 CONCLUSIONS

This paper describes our proposed semi-supervised multimodal emotion recognition method based on self-training for the MER2023-SEMI Challenge. It leverages the multimodal prediction consistency to reduce the confirmation bias, and utilizes the class-balanced top-k strategy to select the high-confidence pseudo-labeled samples. The proposed method is validated to be effective, with the weighted F1 score reaching 88.53% on the test set of the MER2023-SEMI Challenge. In future work, we intend to consider the regression problem in SSL for utilizing the pseudo labels of dimensional emotion, and further enhance the performance by integrating additional semi-supervised learning techniques such as Consistency Regularization.

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